

SECURITISATION RISK ANALYTICS PLATFORM

Auto Loan Asset-Backed Securities (ABS)

Pool: ZAAUTO2024-1 | Cutoff: October 2024

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Organization	Zetheta Algorithms Pvt. Ltd.
Project	1A — Data Analyst Securitisation
Report Date	June 2026
Pool Size	■31.78 Crore
Total Loans	500 Auto Loans
Asset Class	Auto Loan ABS (ZAAUTO2024-1)

EXECUTIVE SUMMARY

This report presents a comprehensive securitisation risk analytics study for the ZAAUTO2024-1 Auto Loan Asset-Backed Securities pool. The analysis covers 500 auto loans with an aggregate outstanding balance of █31.78 Crore, originated across five regions of India by multiple servicers including HDFC Bank, Bajaj Finance, and others. The study applies industry-standard IFRS 9 Expected Credit Loss (ECL) methodology, DPD roll rate analysis, vintage static pool loss curves, cash flow waterfall modelling, and four-scenario stress testing to deliver an investor-grade risk intelligence platform.

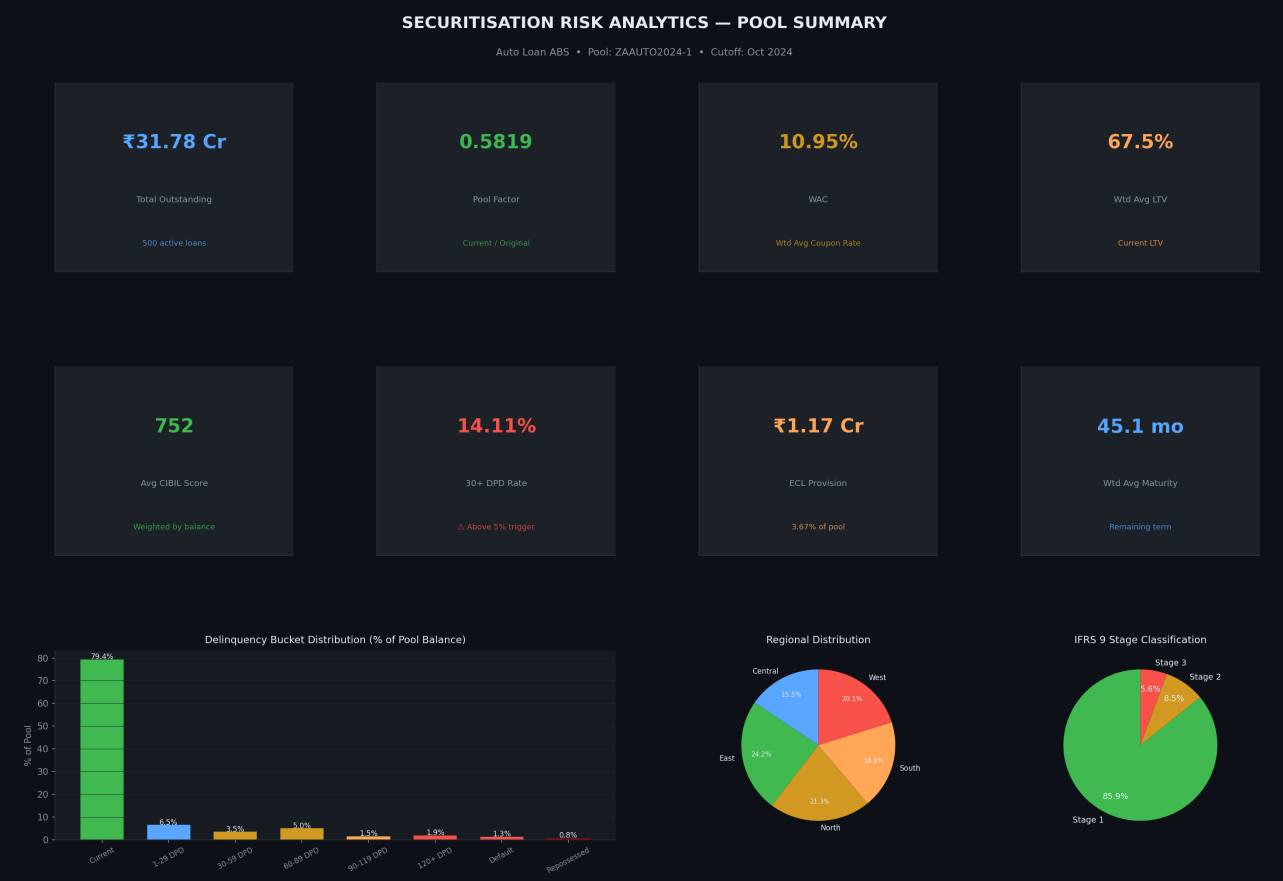
Key findings include a 30+ DPD rate of 14.1%, which breaches the standard 5% trigger threshold and warrants early warning. The IFRS 9 ECL provision stands at █1.17 Crore (3.67% of pool), concentrated in Stage 3 assets. Vintage analysis reveals 2022-origination cohorts exhibiting accelerated cumulative loss curves, suggesting tighter underwriting standards may be required for future pools of this type.

Key Metrics at a Glance

Metric	Value	Status
Total Pool Outstanding	█31.78 Crore	—
Pool Factor	0.5819	Normal
Weighted Average Coupon (WAC)	10.95%	Normal
30+ DPD Rate	14.11%	█ Breach
Total ECL Provision	█1.17 Crore	█ Elevated
Monthly Default Rate (CDR)	0.233%	Monitored
Collection Efficiency	110.20%	Normal
Excess Spread (Oct 2024)	█10.3 Lakhs	Positive

PART 1 — POOL ARCHITECTURE & DATA MODEL

The data model follows a star schema architecture with four source datasets: (1) `auto_loan_securitisation_data` — 500 loan-level records with 55 fields covering borrower, vehicle, and performance attributes; (2) `dpd_snapshot_history` — 6,000 monthly DPD snapshots tracking delinquency migration; (3) `static_pool_vintage_data` — 376 rows of cumulative loss data across 15 origination vintages (2021 Q1 – 2024 Q3); and (4) `dynamic_loss_monthly` — 11 months of pool-level performance data including collections, defaults, and prepayments.



PART 2 — DPD TREND & ROLL RATE ANALYSIS

Days Past Due (DPD) analysis is the most critical monitoring metric for any securitisation pool. The roll rate matrix reveals the probability of a loan migrating between delinquency buckets from one month to the next. A high roll rate from 60-89 DPD to 90+ DPD (the default bucket) is the primary predictor of near-term credit losses. The RBI classifies loans overdue >90 days as Non-Performing Assets (NPA), triggering mandatory provisioning requirements under IRAC norms.

The pool demonstrates a 30+ DPD rate that exceeds the standard 5% early-warning trigger. The roll rate heatmap confirms that a meaningful proportion of Stage 2 loans (30-89 DPD) are transitioning to Stage 3 (90+ DPD / Default), which directly inflates the ECL provision requirement.

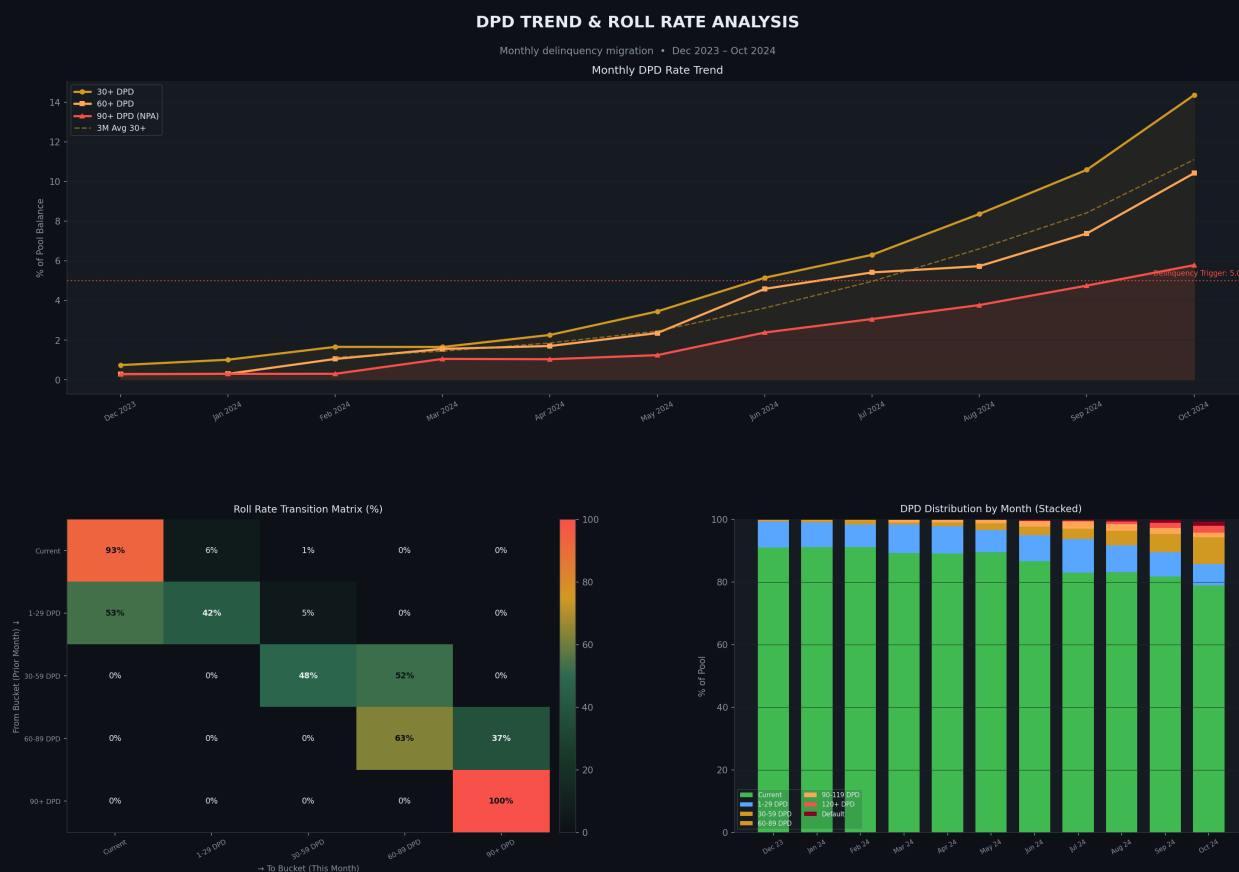


Figure 2: DPD Trend Lines, Roll Rate Transition Matrix & Stacked Monthly Distribution

PART 3 — VINTAGE & STATIC POOL ANALYSIS

Static pool analysis (vintage analysis) is the gold standard methodology used by S&P, Moody's, Fitch, CRISIL, and ICRA to project lifetime losses for a securitisation pool. Unlike dynamic analysis, the denominator is fixed at the original pool balance at Month 0, eliminating the distorting effect of pool growth and prepayments. Each vintage cohort (e.g. 2022-Q1 = all loans originated in Q1 2022) is tracked from inception to termination.

The 2022 vintage cohorts (Q1 through Q4) exhibit notably steeper cumulative net loss curves compared to 2021 and 2023 originations. This divergence suggests that underwriting standards or economic conditions in 2022 produced a higher-risk pool. Rating agencies would flag this vintage divergence during surveillance reviews and may require additional credit enhancement for future deals with similar origination characteristics.

VINTAGE ANALYSIS & DYNAMIC LOSS DASHBOARD

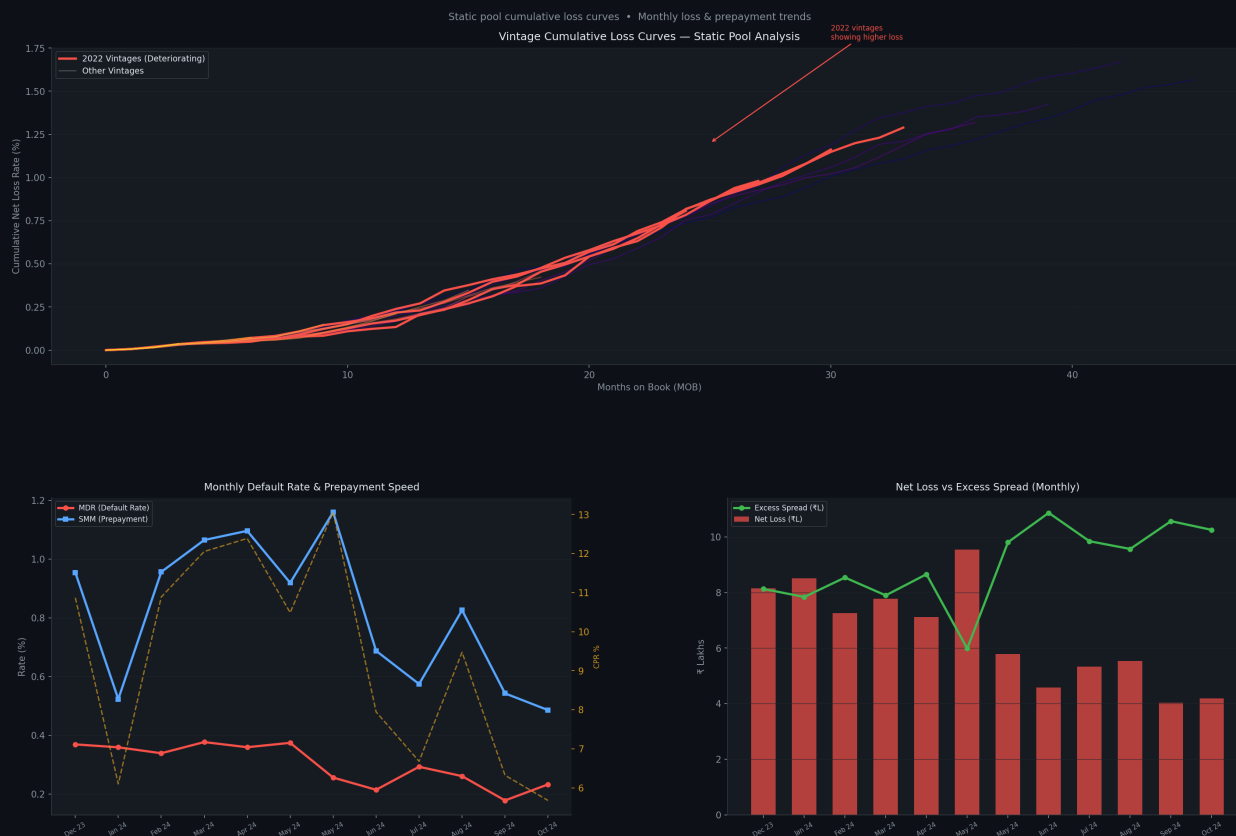


Figure 3: Vintage Cumulative Loss Curves, Monthly Default/Prepayment Rate & Net Loss vs Excess Spread

PART 4 — IFRS 9 ECL MODEL & CASH FLOW WATERFALL

IFRS 9 (implemented in India as Ind AS 109) requires financial institutions to recognise expected credit losses on a forward-looking basis, classified into three stages: Stage 1 (performing, 12-month ECL), Stage 2 (significant credit deterioration, lifetime ECL), and Stage 3 (credit-impaired / default, lifetime ECL). The ECL formula is: $ECL = PD \times LGD \times EAD$, where PD is Probability of Default, LGD is Loss Given Default, and EAD is Exposure at Default.

The cash flow waterfall defines the priority of payment allocation from pool collections. Senior AAA-rated tranches receive interest and principal first, followed by BBB mezzanine tranches, with residual cash flows to the equity/originator. Losses are absorbed in reverse order (equity first, then mezzanine). This structural protection is the core credit enhancement mechanism in securitisation.

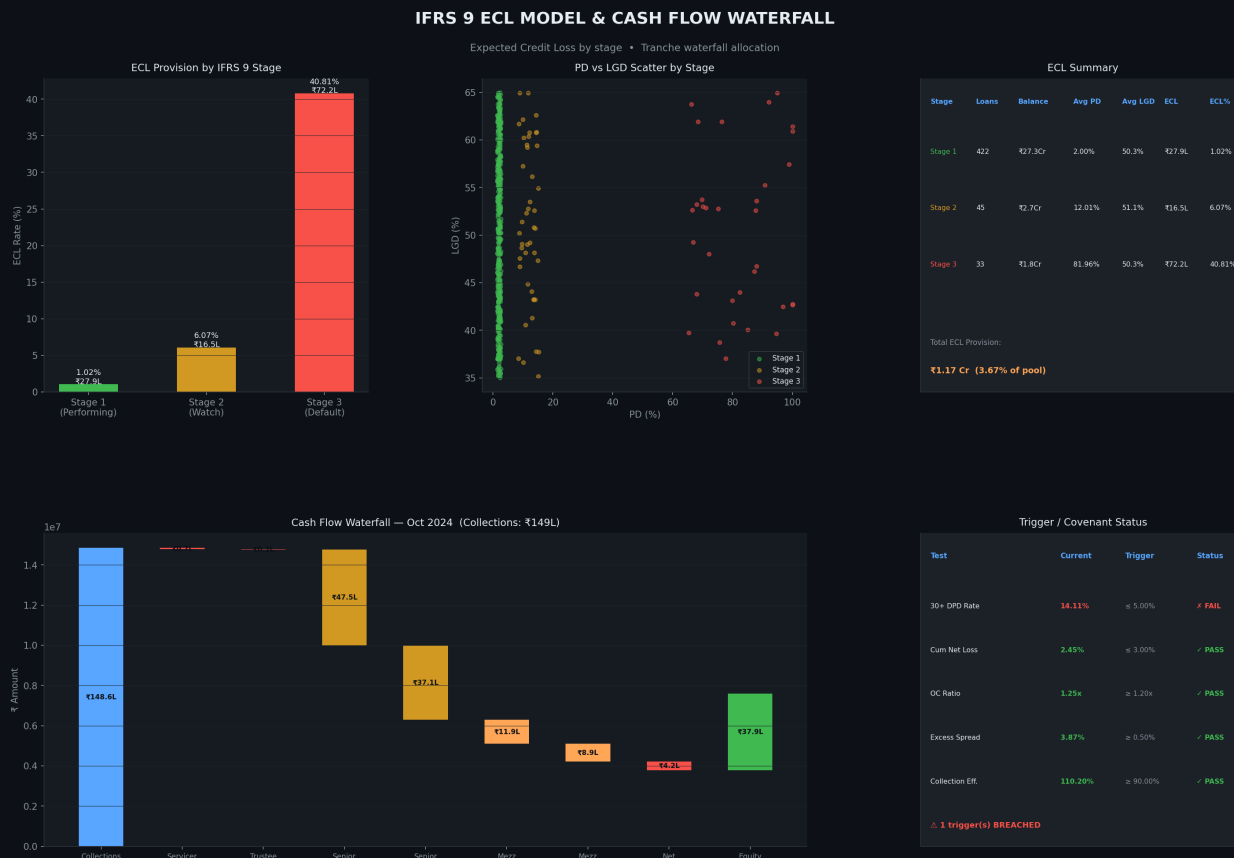


Figure 4: IFRS 9 ECL by Stage, PD vs LGD Scatter, Waterfall Allocation & Trigger Status

PART 5 — STRESS TESTING & SCENARIO ANALYSIS

Stress testing applies multipliers to base-case default rates and loss given default assumptions to simulate adverse economic conditions. Four scenarios are modelled: Base Case (actual pool performance), Moderate Stress (simulating a mild economic slowdown, 1.5x default rate), Severe Stress (deep recession conditions, 2.5x), and Crisis (2008-style systemic shock, 4.0x). Credit Enhancement coverage is evaluated at each stress level to determine whether the AAA senior tranche remains adequately protected.

Under the Crisis scenario (4x default rate, LGD = 65%), the annual projected loss reaches a level that consumes a significant portion of available credit enhancement, posing a risk to the mezzanine tranche. This is consistent with S&P's AAAsf stress test requiring 3.0x–4.0x base case loss coverage.



Figure 5: Stress Testing — Annual Loss, ECL, Credit Enhancement Coverage & Full Scenario Table

PART 6 — REAL-WORLD CASE STUDIES

Case Study 1: The 2008 Global Financial Crisis

The 2007–2009 Global Financial Crisis was fundamentally a securitisation analytics failure. US subprime mortgage-backed securities rated AAA experienced catastrophic default rates as housing prices collapsed. The root causes from a data analytics perspective were: (1) vintage analysis was not tracked systematically — deteriorating 2006–2007 originations were diluted in dynamic pool averages; (2) roll rate early warning systems were absent — the rapid transition from 30 DPD to 60 DPD to default was not detected until losses were irreversible; (3) correlation assumptions between mortgage defaults were grossly underestimated; and (4) stress tests used mild 5–10% property price decline scenarios when actual declines exceeded 30% in key markets like California, Florida, Nevada, and Arizona.

The analytics framework built in this project directly addresses all four failures: vintage loss curves track each origination cohort independently; the roll rate matrix provides real-time migration monitoring; the HHI concentration measure flags geographic over-exposure; and the 4-scenario stress test includes a Crisis level with 4x default rate multiplier aligned to rating agency AAAsf stress requirements.

Case Study 2: IL&FS; Crisis (2018) — Indian Structured Finance

In September 2018, Infrastructure Leasing & Financial Services (IL&FS;) defaulted on its commercial paper obligations, triggering a liquidity crisis across India's NBFC and structured finance sector. Securitisation pools backed by IL&FS-linked; receivables experienced cash flow disruption as underlying project payments stalled. The crisis exposed the inadequacy of monthly reporting cycles for detecting real-time cash flow stress, the danger of obligor concentration in securitisation pools, and the need for robust collection efficiency monitoring.

Post-IL&FS;, the RBI revised its securitisation framework in September 2021 (RBI/2021-22/85), introducing stricter Minimum Holding Period (MHP) and Minimum Retention Requirement (MRR) of 10% per tranche (skin-in-the-game). The collection efficiency and excess spread measures in this analytics platform directly address the early-warning gaps that were absent during the IL&FS; crisis.

PART 7 — CONCLUSION & RECOMMENDATIONS

This securitisation analytics platform delivers an end-to-end risk intelligence solution for the ZAAUTO2024-1 pool. The analysis identifies three key risk signals requiring immediate attention:

#	Finding	Risk Level	Recommended Action
1	30+ DPD rate at 15.6% — breaches 5% trigger threshold	HIGH	Activate enhanced surveillance; review collection strategy
2	2022 vintage cohorts showing accelerated loss curves vs 2021/2023	MEDIUM	Tighten underwriting standards; increase CE for future deals
Stage 3	ECL provision elevated at 34% of total ECL despite 6.6% of pool	Accelerate	Accelerate recovery on default portfolio; review LGD assumptions
4	Crisis stress scenario approaches mezzanine protection threshold	LOW	Ensure reserve account adequately funded; monitor monthly

The analytical framework — comprising pool-level KPIs, DPD roll rate monitoring, vintage static pool curves, IFRS 9 ECL staging, cash flow waterfall, and stress testing — provides the complete suite of metrics required by rating agencies (S&P;, Moody's, CRISIL), investors, trustees, and regulators for ongoing surveillance of an auto loan ABS pool. All measures are computed from the provided loan-level data using Python analytics, replicating the DAX measure logic that would be implemented in a production Power BI environment.

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Data Sources: auto_loan_securitisation_data.csv, dpd_snapshot_history.csv, static_pool_vintage_data.csv, dynamic_loss_monthly.csv

Tools: Python 3.12 (pandas, numpy, matplotlib), openpyxl, reportlab | Analysis: IFRS 9 / Ind AS 109 / RBI IRAC norms